Part II: Philosophy Module

Marks: 40
Approx. Time: 45 minutes

Section A: **5 x 2**

1. State whether the following arguments are inductive or deductive. Also identify the type of inductive and deductive arguments they are (in terms of the labels that apply to each type in terms of sound/unsound/weak/strong/valid/invalid).

RUBRIC: 1 mark for getting inductive/deductive right; 1 for getting the label that applies to it right; No marks for getting the first part wrong; 1 for getting the first right and the second label wrong.

(a) The same pattern of questions was followed in the last two years' question papers. This year too, the pattern will be similar.

- **Inductive, weak**

(b) The highway sign says that the risk of forest fires in this area is extremely high right now. Therefore, we must conclude that the risk of forest fires is indeed high right now.

- **Inductive, strong**

(c) If people are not informed about the workings of government, then they cannot vote intelligently. Since people cannot vote intelligently, it is true that that people are not informed about the workings of government.

- **Deductive, invalid ('modus ponens, fallacy of affirming the consequent' is right as well)**

(d) Either the painting is a forgery, or it's worth a small fortune. Therefore, the painting is worth a small fortune, since it's not a forgery.

- **Deductive, valid**

(e) If the Big Bang theory is correct, then the universe is billions of years old. And if the Big Bang theory is correct, then the universe was not created in six days. Thus, if the universe is billions of years old, then it was not created in six days.

- **Deductive, invalid**

Section B:

2. Identify the following informal fallacies. Give reasons for the same. Note that each of these is representative and uniquely identifiable as the instance of a single specific fallacy. Provide the specific fallacy each falls under. Do not write multiple answers for the same question.
5 x 2

RUBRIC: 1 mark for identifying the fallacies correctly; 1 mark for reasoning (just look for whether they've got the gist of the reason right). Zero marks for getting the first part wrong; 1 for getting the first right and the reasoning wrong.

(a) We've all heard the complaint that millions of Indians are without adequate health care. But India's doctors, nurses, and hospitals are among the best in the world. Thousands of people come from abroad every year to be treated here. Clearly there is nothing wrong with our health-care system.

- **Red herrring;**
  - **the argument is about inaccessibility of health care among millions of Indians. But the response is diverting attention away from this and towards the quality of healthcare in India.**

(b) Most people think ad hominem is not a fallacy, so it really doesn't matter if you commit it.

- **Appeal to populace (ad populum)**
  - **the conclusion weighs on the premise that since most people think x, x is permissible.**

(c) There is so much conflicting news these days. It's better to stay away from news altogether.

- **Missing the point**
  - **there is no second premise provided but a jump from the first to the conclusion.**

(d) It is being argued that too much time on the Internet is leading to brain rot in youngsters these days. Yet a lot of content on the Internet is excellent. Youtube has so many informative and educational videos and one can connect with people from across the globe via social media. You can catch up with the latest news from all over the world. The Internet is very resourceful, indeed.

- **Red herring**
  - **the argument is about brain rot in youngsters from consuming too much Internet content. Attention is diverted away from this and towards the resourcefulness of content on the Internet, which may be so, but not relevant to the topic at hand.**

(e) The editors of the Daily Register have accused our company of being one of the city's worst water polluters. But the Daily Register is responsible for much more pollution than we are. After all, they own the Western Paper Company, and that company discharges tons of chemical residue into the city's river every day.

- **Tu quoque/ad hominem circumstantial (either of the two work; 0.5 if they've only written it as ad hominem without identifying it as 'circumstantial')**
  - **the argument rests on the claim 'that they do it too'.**

<u>**Section C:**</u>

3. Convert the following inferences to match with the label in the brackets. You may choose to add new premises and/or modify the premises provided wherever necessary: 10 x 2

<u>**RUBRIC: 1 mark for converting to deductive/inductive; 1 for making it valid (i.e., if the premises are true, then the conclusion has to be true) or strong/weak (i.e., the truth of the premises makes the conclusion more/less probable).**</u>
<u>**There would be variations in how students go about their conversions. Use your logical discretion in evaluating them. If in doubt, pls reach out to me.**</u>

<u>**NOTE: make sure they have not tampered with the conclusion. e.g., they are not allowed to turn 'they must be dating' to 'they must not be dating.' However, they can tweak the same to say 'they are likely to be dating'.**</u>

(a) You see the two of them together always. They must be dating. **(valid, deductive)**

All those who are seen together are those who must be dating
the two of them are seen together
therefore, the two of them must be dating

(b) I've taken that professor's course and noticed that she does not bother those who leave class after attendance. So if you want attendance without having to attend classes, then you can take her course. **(strong, inductive)**

In all the courses that professor has taken, she has not bothered those who left the class after attendance. So <u>it is likely</u> that if you take her class, you may get leave after attendance without having to sit through the class.

(c) Thursday follows Tuesday. Therefore, Hyderabad is the capital of Telangana. **(valid, modus ponens)**

If Thursday follows Tuesday, then Hyderabad is the capital of Telangana.
Thursday follows Tuesday.
Therefore Hyderabad is the capital of Telangana

Make sure this has the structure of a modus ponens:
If p then q; p, therefore q

(d) I have so far not used cash in Hyderabad since all the shops accepted UPI payments. So you don't need to carry any cash in this city. (strong, inductive)

The above inference is weak since it relies solely on personal experience. Turning personal experience into a larger sample will make it more likely and a strong, inductive.

Most of the shops in Hyderabad accept UPI payments, so you will most probably/likely not need to carry any cash in this city.

(e) I've never seen her use a phone. She must be anti-technology **(valid, deductive).**

All those who don’t use a phone are anti-technology
She does not use a phone
therefore, she is anti-technology

(f) I never got flagged for plagiarism when i used that LLM for writing my assignments. It is foolproof. **(strong, inductive).**

Same as (d)

This LLM has been tested to be most effective in not being detected for plagiarism. So if you were to use it, you are highly likely to not get flagged for plagiarism

(g) If there is fuel, the motorcycle will turn on. (sound, modus tollens).

To turn this into a sound if-then argument, the antecedent and consequent must be reversed so that the antecedent becomes the sufficient condition for the consequent and the consequent a necessary condition for the antecedent.

If the motorcycle turns on, then there is fuel
there is no fuel
therefore, the motorcycle will not turn on

<u>Award 1 mark if they get a valid, but not sound, modus tollens:</u>

if there is fuel, the motorcycle will turn on
the motorcycle is not turning on
therefore, there is no fuel

But it should be in the form of modus tollens:
if p, then q; not q; therefore, not p

(h) If there is oxygen, then there is fire. (sound, modus ponens)

<u>Similar to the above, the antecedent and consequent must be interchanged to make it sound:</u>

If there is fire, then there is oxygen
there is fire,
therefore there is oxygen

<u>Award 1 mark if they get a valid, but not sound, modus ponens:</u>

If there is oxygen, then there is fire
there is oxygen,
therefore, there is fire

(i) If you exercise, then you will reduce your chances of arthritis **(strong, inductive)**

<u>Expanding the sample size and making it probable rather than certain will make it into a strong, inductive argument.</u>

Studies on a large number of participants have shown that those who exercise reduce their chances of arthritis. Hence, if you exercise, it is likely that you too will reduce your chances of arthritis.

(j) All animals are mammals. And since Jumbo is a mammal, it is also an animal **(valid, deductive)**

All mammals are animals
Jumbo is a mammal
Therefore, Jumbo is an animal

1. (a) What is philosophy and what is the philosophical method grounded in (2)? What marks a philosophical question as distinct from a scientific question (2)? How has it come to be that philosophy deals almost exclusively with unanswerable questions (2)?

What is philosophy? (1 mark; any one point)

- philosophy is the love of wisdom (philo + sophia).
- Philosophy is a normative discipline which involves clarifying concepts, questioning assumption, defining terms and presenting sound arguments that are grounded in reason.

What is the philosophical method grounded in (1 mark; any 1 point)

- the philosophical method is grounded in logic/sound reasoning.
- Philosophy aims at stating things clearly and precisely.
- The philosophical method uses logic to make arguments that give us good reasons to believe in the conclusion the argument presents.

Philosophical question vs. scientific question
Philosophical question (1 mark for any one point)

- philosophy concerns itself with **"normative" questions—what ought to be the case**
- philosophical question also involves questioning our basic assumptions

Scientific question (1 mark for any one point)

- A scientific question is **descriptive**. It has to do with **what is the case.**
- Scientific question deals with facts/state of affairs

How come philosophy deals with unanswerable questions (2 marks; 1 mark each; any one point below)

- All disciplines at one time or another were part of philosophy and as answers to foundational questions appeared, they **spurred new disciplines, while the unanswered questions remained within philosophy.** For example, science was called natural philosophy. Once the scientific method was discovered, it went on to become a discipline in its own right (example is optional; 0.5 marks if they've only provided example).
- Since **philosophy deals with normative questions which cannot be settled using the scientific method**, it seems to be that the questions that philosophy takes up are never settled always up for debate and hence unanswerable.

(b) Briefly describe the three laws of logic (3). Bring out the difference between logical impossibility and physical impossibility (2). Give examples that match the following descriptions:

(i) logically possible but physically impossible (1)
(ii) logically impossible but physically possible (1)

<u>three laws of logic (1 mark each; 0.5 marks for the label + 0.5 marks for the description; total 3 marks)</u>

- **Law of noncontradiction**
  - something cannot both be A and not-A at the same time
    - alternatively, Nothing can both have a property and lack it at the same time
- **Law of identity**
  - everything is identical to itself

        - alternatively, A = A
- **law of excluded middle**
    - Either A or not-A
        - for any property, everything either has it or lacks it

logical impossibility (1)

- anything which is inconsistent with the laws of logic is logically impossible,

physical impossibility (1)

- whereas anything which goes against the laws of physics is physically impossible.

examples that match the following descriptions:

(i) logically possible but physically impossible (any one; 1 mark)

- Archimedes saying, “Give me a lever long enough and a fulcrum on which to place it, and I shall move the world” represents a physical impossibility but a logical possibility.
- The possibility of a human jumping to the moon from the Earth
- Running 100 meters in 1 second.
- Traveling faster than the speed of light.
- A cow jumping over the moon

(ii) logically impossible but physically possible (any one; 1 mark)

- quantum mechanics as putting to question the law of non-contradiction.
- Give the benefit of doubt to time travel.

(c) Outline any 2 features that distinguish inductive from deductive arguments (3). What is the principle/ground upon which inductive and deductive inferences are founded (2)? How is it that the grounds of inductive inference involves a form of circular argument (2)?

<u>Inductive vs. deductive arguments (1.5 MARKS FOR ANY 2 OF THE FOLLOWING POINTS)</u>

(1) In inductive arguments, the truth of the premises makes the conclusion more or less likely/probable, whereas in a deductive argument, the truth of the premises guarantees the truth of the conclusion.
(2) Inductive arguments are either strong or weak, whereas deductive arguments are either valid or invalid/sound or unsound.
(3) Inductive arguments rely on sense experience (a posteriori), whereas deductive arguments rely on rationality (a priori).
(4) Inductive arguments are NOT truth-preserving, whereas deductive arguments are truth-preserving (i.e., if the premises are true in a valid deductive argument, the conclusion HAS TO BE true).
(5) additional premises can make a strong inductive argument weak (and vice-versa), whereas additional premises cannot make a valid deductive argument invalid.
(6) inductive arguments give us only probability, whereas deductive arguments give us certainty.

What is the principle/ground upon which inductive and deductive inferences are founded (2)?

The principle/ground upon which inductive inferences are founded (1 mark)

- the **principle of the uniformity of nature** which tells us that the future will be like the past is what grounds inductive inferences.

The principle/ground upon which deductive inferences are founded (1 mark; any 1 point)

- deductive inferences are grounded on the principle that the truth of the conclusion is already contained within the truth of the premises.
- They are grounded in the notion of logical impossibility which prevents true premises from leading to a false conclusion
- grounded in the notion of truth-preservation in that if the premises are true, the conclusion has to be true

Why does this make it an instance of a circular argument (2 marks; award 2 marks if they've captured the essence of the point below)

- Inductive inferences are grounded in the principle of the uniformity of nature (or that the future will resemble the past). The principle of uniformity of nature (or that the future will resemble the past) is itself an inductive inference, thus making it an instance of a circular argument.